SIMATS ENGINEERING

Saveetha School of Engineering

ASSESSMENT TOOL 1

Analytical Problem Solving

CSA15 · Cloud Computing and Big Data Analytics

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments (BL3)

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Code of Conduct: I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

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> **KEY INSIGHT How to read this report**
> Each answer follows one pattern: **Problem Interpretation → Analysis (cause–effect) → Method Selection → Solution → Key Insight**, mirroring the five rubric criteria of the Level 4 (Exemplary) band.

1. Type-1 vs Type-2 Hypervisors for Mission-Critical Clouds

1.1 Problem Interpretation

A datacenter hosting mission-critical applications needs **near-native performance, strong tenant isolation, high availability and scalable management**. The task is to compare the two hypervisor classes on performance, security and manageability and justify one.

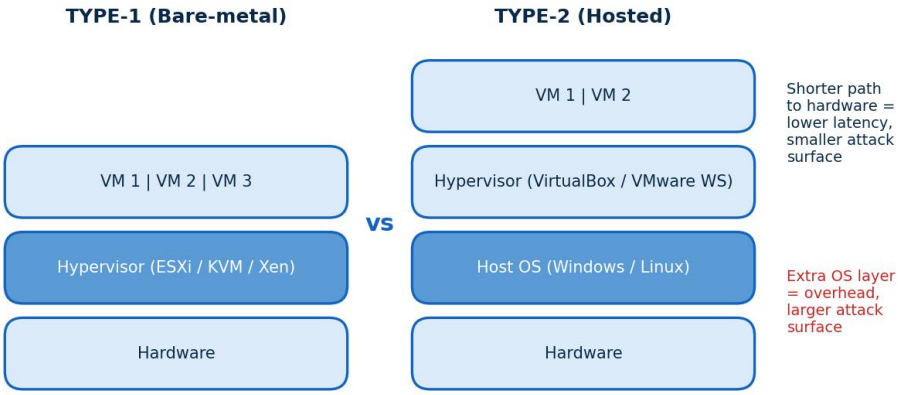


Figure 1: Type-1 (bare-metal) vs Type-2 (hosted) hypervisor stacks

1.2 Comparative Analysis

Criteria	Type-1 (Bare-metal)	Type-2 (Hosted)
Performance	Runs directly on hardware; CPU/memory/I/O reach VMs with ~1–5% overhead; supports VT-x/EPT, SR-IOV, NUMA awareness.	Requests pass through the host OS scheduler and drivers; 10–30% overhead and unpredictable latency from host processes.
Security	Small trusted computing base (TCB); no general-purpose OS; fewer patches and services to attack.	Large TCB: any host-OS vulnerability or malware compromises every VM.
Manageability	Cluster managers (vCenter, OpenStack, Proxmox): live migration, HA, DRS, API automation.	Per-machine desktop tools; no native clustering, HA or large-scale orchestration.
Scalability	Hundreds of VMs per host; multi-host clusters.	Limited to a handful of VMs per workstation.
Examples	VMware ESXi, KVM, Xen, Microsoft Hyper-V (used by AWS Nitro, Azure, GCP).	Oracle VirtualBox, VMware Workstation, Parallels.
Best fit	Production cloud, enterprise datacenter.	Development, testing, training labs.

1.3 Justified Decision

Type-1 is the better choice. Mission-critical workloads need predictable latency (performance), minimal attack surface (security) and features such as live migration, automated failover and policy-based management (manageability). Only Type-1 provides all three; hyperscalers such as AWS (Nitro/KVM) and Azure (Hyper-V) run exclusively on Type-1 designs. Type-2 remains valid only for desktops and test labs.

> **KEY INSIGHT Why it matters**
> Every extra software layer adds latency **and** attack surface – Type-1 removes the host OS layer, improving both at once.

2. Workload Isolation and Attack Surfaces

2.1 Problem Interpretation

Finance, healthcare and student-portal workloads share the same physical hardware but carry different compliance duties (PCI-DSS, HIPAA, FERPA/DPDP). The provider must guarantee that **no tenant can read, alter or disrupt another** and must identify the main ways that guarantee can be broken.

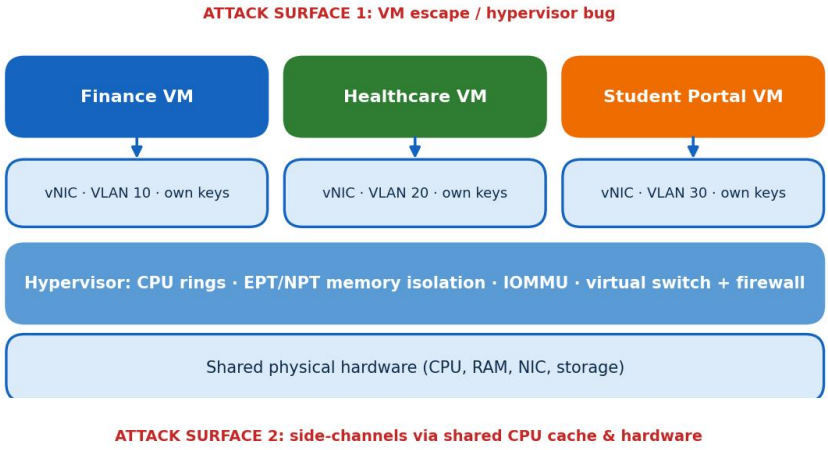


Figure 2: Multi-tenant isolation layers and the two major attack surfaces

2.2 How Virtualization Provides Isolation

Resource	Isolation Mechanism	Benefit
CPU	Hardware privilege rings; VT-x / AMD-V root vs non-root mode	Guest code cannot execute privileged instructions on the host
Memory	Separate address spaces via EPT / NPT; page-sharing disabled for sensitive VMs	One VM cannot read another's RAM
I/O devices	IOMMU (VT-d) with DMA remapping; virtio / SR-IOV	Blocks rogue-device DMA attacks
Network	Per-tenant vNICs, VLAN/VXLAN, virtual firewall, micro-segmentation	Traffic separated even on shared switches
Storage	Per-tenant virtual disks, LUN masking, encryption with tenant-owned keys	Data unreadable across tenants

Additional policy-level controls: **dedicated hosts** (affinity / anti-affinity rules) for finance and healthcare, separate management VLANs, role-based access and per-tenant audit logs.

2.3 Two Major Attack Surfaces

Attack Surface	Threat and Real Example	Mitigation
① Hypervisor / VM escape	A bug in emulated devices lets a malicious guest run code on the host and reach all tenants (e.g. VENOM, CVE-2015-3456, floppy-controller flaw).	Patch quickly; remove unused virtual devices; sVirt / SELinux; minimal hypervisor; secure management plane with MFA.
② Shared-hardware side-channels	Co-resident VMs infer secrets from cache or speculative execution timing (Spectre, Meltdown, L1TF).	Dedicated hosts for regulated tenants; core scheduling / SMT off; microcode and kernel mitigations; confidential computing (AMD SEV, Intel TDX).

KEY INSIGHT Insight
Logical isolation is only as strong as the hypervisor beneath it – so regulated workloads should combine **VM isolation + dedicated hosts + encryption**.

3. VM Host Utilization Analysis

3.1 Problem Interpretation

Snapshot: **CPU 92%, memory 88%, storage I/O latency 35 ms, VM boot time 170 s.** Using common operating benchmarks (CPU/memory ≤ 80%, SSD latency ≤ 10 ms, boot ≤ 60 s), every metric is outside its healthy range.

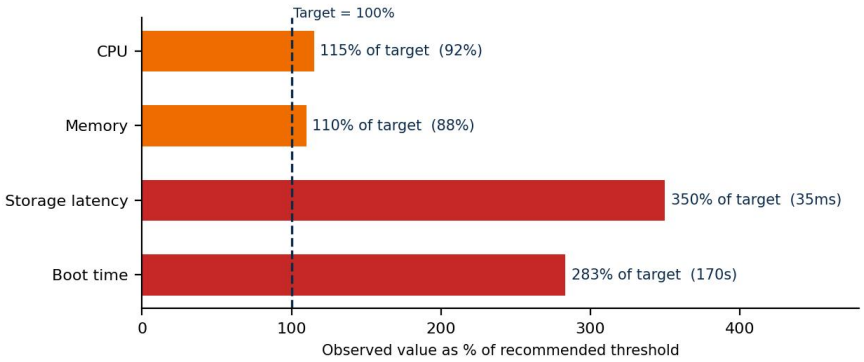


Figure 3: Observed metrics versus recommended thresholds (100% = target)

Metric	Observed	Healthy Target	Diagnosis
CPU	92%	≤ 80%	Saturated; high CPU-ready time from vCPU over-commitment
Memory	88%	≤ 80%	Near ballooning/swapping threshold; risk of host swap
Storage latency	35 ms	≤ 10 ms (SSD)	3.5× target – primary bottleneck
VM boot time	170 s	≤ 60 s	Symptom of disk contention and CPU wait

3.2 Likely Bottlenecks – Cause and Effect

- **Storage I/O is the root bottleneck.** Booting reads OS images, drivers and logs at once; with 35 ms latency each read stalls, so boot time stretches to 170 s (a boot storm).
- **Memory pressure amplifies it.** At 88%, the host begins ballooning and swapping to the same slow disk, raising latency further – a vicious cycle.
- **CPU saturation delays everything.** At 92%, vCPUs queue for physical cores, slowing I/O completion handling and guest start-up.

3.3 Corrective Actions

Priority	Action	Expected Result

Priority	Action	Expected Result
1 – Storage	Move VM disks to NVMe/SSD tier; add host-side cache; use thin clones from a golden image; enable virtio/paravirtual drivers; separate boot and data datastores.	Latency < 10 ms; boot < 60 s
2 – Memory	Right-size over-allocated VMs; enable ballooning and transparent page sharing; add RAM or migrate low-priority VMs.	Memory ≤ 75%; no host swap
3 – CPU	Live-migrate VMs to idle hosts (DRS); reduce vCPU:pCore ratio to ≤ 4:1; set CPU reservations for critical VMs.	CPU ≤ 75%
4 – Operations	Stagger boots; auto-scaling; monitoring alerts at 80% (Prometheus/CloudWatch).	Prevent recurrence

KEY INSIGHT Insight
Fix **storage first**: it is the common cause behind slow boots and part of the memory-swap loop, giving the largest improvement for the lowest cost.

4. Software Defined Datacenter (SDDC) vs Traditional Datacenter

4.1 Problem Interpretation

The question asks how an SDDC improves **agility** – the speed of provisioning, scaling and changing infrastructure – and how compute, storage and network virtualization each contribute.

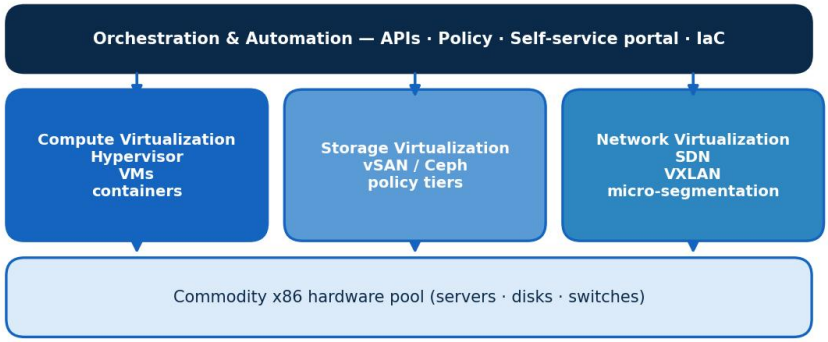


Figure 4: SDDC layered architecture

4.2 Agility Comparison

Dimension	Traditional Datacenter	Software Defined Datacenter
Provisioning time	Days to weeks (tickets, racking, cabling)	Minutes via self-service portal and API
Hardware	Proprietary, siloed per function	Commodity x86 pooled and abstracted
Configuration	Manual, device-by-device; error-prone	Policy-driven, Infrastructure as Code
Scaling	Over-provision for peak	Elastic, automatic scale-out/in
Resource utilization	Typically 10–20%	60–80% through consolidation
Resilience	Hardware-dependent failover	Software HA, live migration, self-healing
Cost model	High CAPEX, vendor lock-in	Lower CAPEX; pay-per-use OPEX

4.3 Virtualization Considerations

Layer	Technology	Agility Contribution
Compute	Hypervisors (KVM, ESXi), containers; templates and auto-scaling	VM in seconds; consolidation; live migration without downtime
Storage	Software-defined storage (vSAN, Ceph); policy tiers; thin provisioning	Capacity added by adding nodes; QoS per VM; no manual LUN carving
Network	SDN controller, OpenFlow, VXLAN overlays, NFV, micro-segmentation	Virtual networks and firewalls created by API; no re-cabling

Real-world example: AWS VPC, Azure Virtual Network and VMware Cloud Foundation (vSphere + vSAN + NSX) deploy complete multi-tier environments in minutes using Terraform or CloudFormation templates.

> **KEY INSIGHT Insight**
> Agility comes from **abstraction + automation**: when all three resource types are software-defined, a whole environment becomes a reusable template.

5. Multi-Cloud Identity Federation: Failure Analysis and Design

5.1 Problem Interpretation

The federation fails because each provider keeps its own identity silo. A user is recognised in one cloud but not in another, so authentication or authorisation breaks.

5.2 Root-Cause Analysis

Cause	Effect
Different identity stores and formats (e.g. email vs UPN vs ARN)	Same user maps to different, unmatched identifiers
Mixed protocols (SAML 2.0, OIDC, proprietary tokens)	Tokens are rejected or cannot be parsed
No common trust relationship or metadata exchange	Assertions are unsigned or untrusted
Inconsistent roles/attributes and manual provisioning	Orphaned accounts, over- or under-privileged access
Over-sharing of personal attributes in tokens	Privacy and compliance risk (GDPR / DPDP Act)

5.3 Proposed Secure Identity and Privacy Design

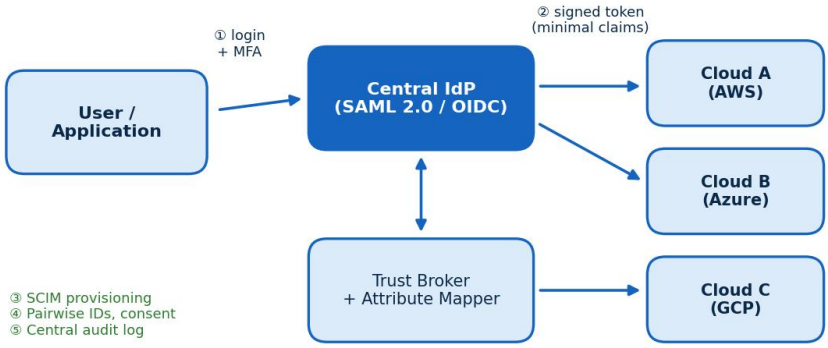


Figure 5: Federated identity architecture across clouds

Design Element	Solution
Trust model	One authoritative IdP (e.g. Microsoft Entra ID / Keycloak); each cloud is a Service Provider with signed metadata exchange
Standards	SAML 2.0 for enterprise SSO, OpenID Connect / OAuth 2.0 for APIs; a broker translates between them
Identity mapping	Canonical attribute schema; trust broker maps local IDs; SCIM 2.0 for automated provisioning and de-provisioning
Access control	Role/attribute-based (RBAC/ABAC) using federated claims; least privilege; MFA on all logins
Token security	Signed (RS256) short-lived tokens; TLS 1.3; audience restriction; clock-synchronised (NTP)
Privacy	Pairwise pseudonymous identifiers, data minimisation (release only needed claims), user consent, no PII in logs
Governance	Central audit trail, periodic access reviews, key rotation, compliance with GDPR/DPDP

KEY INSIGHT Insight
Federation succeeds when **trust is established once at the IdP** and identities are translated by standard protocols – not re-created in every cloud.

6. Conclusion

This analysis shows that cloud performance, security and agility are all determined by virtualization design choices:

- **Type-1 hypervisors** give mission-critical clouds the best performance, security and manageability.
- **Multi-tenant isolation** relies on hardware-assisted virtualization, but the hypervisor and shared hardware remain the key attack surfaces.
- **Capacity problems** should be diagnosed by cause and effect – storage latency first – before adding hardware.
- **SDDC** replaces manual, siloed infrastructure with automated, policy-driven pools.
- **Federated identity** with a trusted IdP, standard protocols and privacy-by-design enables secure multi-cloud operation.

Rubric Alignment (Level 4 – Exemplary)

Criteria	How It Is Met
Problem Interpretation (20%)	Each answer opens with an interpretation of all stated technical requirements
Analytical Reasoning (25%)	Explicit cause–effect chains (Q3 storage–memory–CPU loop; Q5 root-cause table)
Method Selection (20%)	Appropriate concepts: VT-x/EPT, IOMMU, SDN, SDS, SAML/OIDC, SCIM, RBAC
Solution Quality (20%)	Feasible, measurable actions and justified decisions with real cloud examples
Presentation (15%)	Numbered answers, tables, diagrams and key-insight callouts

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